TSA Mock 2

# Problem Solving

The hardest past TSA Problem Solving questions not in Mock 1, any type, from all 16 papers (2008–2023)

| Questions | Total marks | Time limit |
|---|---|---|
| 25 | 25 | 45 minutes |

## Instructions

- You have **45 minutes** to answer all **25 questions**.
- Each question is worth **one mark**. There are no penalties for incorrect answers, so attempt every question.
- Each question has five options, A to E. Choose the one option you consider correct.
- Calculators are **not** permitted. You may use this paper for rough working.
- Every question is a real past TSA question. The year and number in brackets show where it comes from, e.g. **7 (2019 Q32)**.

## What is in this paper

The 25 Problem Solving questions judged hardest of all those not already in Mock 1, from any question type. Some have a debatable official answer: the mark scheme uses the official answers, and the worked solutions explain the debate.

## How the time limit was set

The time limit follows the TARA Problem Solving module (22 questions in 40 minutes), scaled to 25 questions: 25 × 40 ÷ 22 = 45.45 minutes, rounded to the nearest minute (45 minutes). These are hard questions, so expect to run short of time.

**Score: _____ / 25**

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**1** (2015 Q30) Towards the end of a sports season in which all teams in the league play each other four times, the league positions are as follows:

| | **Played** | **Won** | **Drawn** | **Lost** | **Pts** |
|---|---|---|---|---|---|
| East Rovers | 16 | 8 | 7 | 1 | 31 |
| West Athletic | 14 | 7 | 3 | 4 | 24 |
| North Rangers | 17 | 6 | 4 | 7 | 22 |
| South United | 17 | 4 | 10 | 3 | 22 |
| Top Town | 13 | 3 | 8 | 2 | 17 |
| Bottom Albion | 15 | 3 | 6 | 6 | 15 |

Three points are awarded for a win, 1 point per draw and 0 for a loss. At the end of the season, if two teams have the same number of points, the league will be decided by the highest number of games won.

How many of the competing teams could win the league when all the games have been completed?

**A** 1

**B** 2

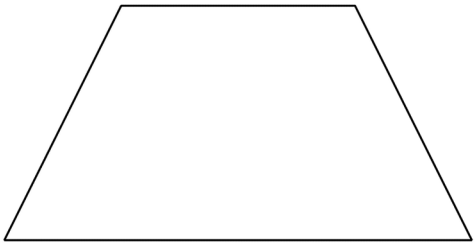
**C** 3

**D** 4

**E** 5

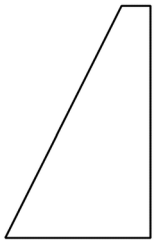
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**2** The trapezium below may be cut into four identical pieces.
(2012 Q32)

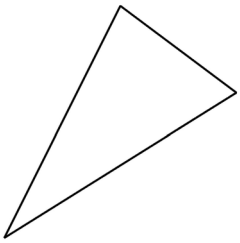


Which of the following pieces can be used four times (flips and rotations are allowed) to make up the trapezium?

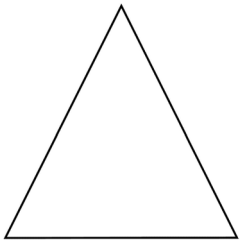
**A**



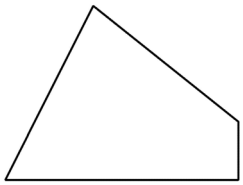
**B**



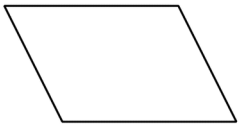
**C**



**D**

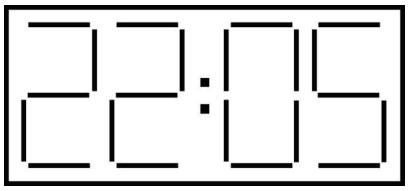


**E**



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**3** (2015 Q38) I stayed in a hotel last night. My room was at the front of the building. At one point I looked out of the window and saw:

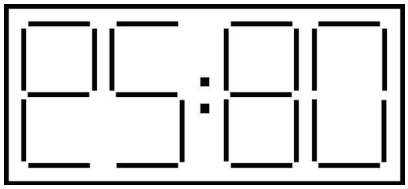



on an illuminated digital clock on the other side of the road. I took the opportunity to reset my watch, which had stopped sometime earlier.

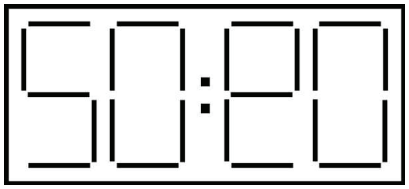
This morning, when I looked out, I realised that what I could see across the road was a reflection in the building opposite of a clock outside the hotel. According to my watch (which I now knew I had reset incorrectly last night) the time was supposedly nine minutes past ten.

What did I see across the road when I looked out this morning?

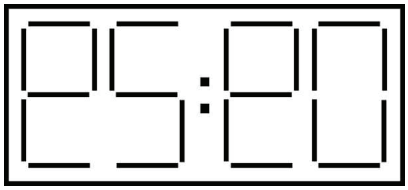
**A**



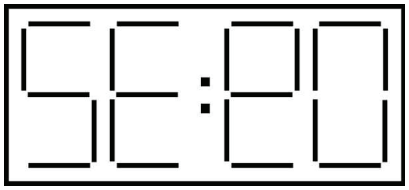

**B**



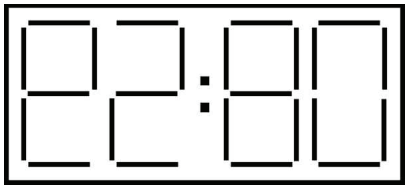

**C**




**D**




**E**




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**4** (2013 Q43) There is a half-hourly train service that operates along the Rowbina Valley, between Erd and Teovil. Leaving Erd at 10 and 40 minutes past each hour, trains stop at Aregon, Lowley, Ergen, Uble and Gidion before arriving at Teovil 51 minutes after departure from Erd. In the opposite direction, trains leave Teovil at 20 and 50 minutes past each hour, arriving at Erd 51 minutes later.

Distances Chart (km)

| | Erd | Aregon | Lowley | Ergen | Uble | Gidion | Teovil |
|---|---|---|---|---|---|---|---|
| Aregon | 6 | | | | | | |
| Lowley | 11 | 5 | | | | | |
| Ergen | 17 | 11 | 6 | | | | |
| Uble | 24 | 18 | 13 | 7 | | | |
| Gidion | 29 | 23 | 18 | 12 | 5 | | |
| Teovil | 36 | 30 | 25 | 19 | 12 | 7 | |

The trains normally average a speed of 60 km/h, but this rises to 80 km/h if they are running late. They stop for three minutes at each station en route, but this is reduced to two minutes when running late.

Yesterday the 10.20 am train from Teovil arrived at Uble on time, but because of a technical fault, remained at Uble for 22 minutes. There were no further delays.

How many minutes late was yesterday's 10.20 am train from Teovil when it arrived at Erd?

**A** 5 mins

**B** 8 mins

**C** 10 mins

**D** 13 mins

**E** 15 mins

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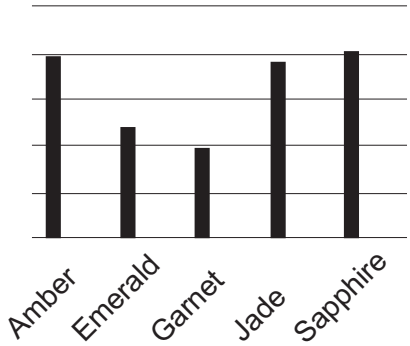
**5** (2020 Q50) The table below shows the medal count by the top five teams in a recent Mini Olympics.

| *teams* | *gold medals* | *silver medals* | *bronze medals* |
|---|---|---|---|
| Amber | 8 | 7 | 6 |
| Emerald | 5 | 10 | 6 |
| Garnet | 6 | 6 | 10 |
| Jade | 6 | 5 | 5 |
| Sapphire | 11 | 8 | 9 |

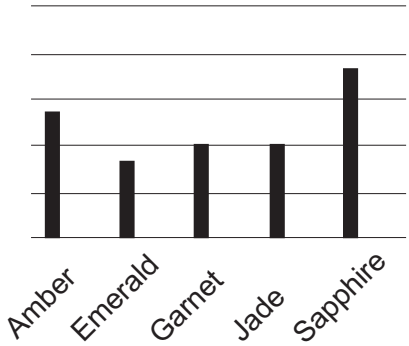
Each team calculated the proportion of their medals that were gold.

Which one of the following charts, with appropriate labelling, shows this proportion for each of the five teams?

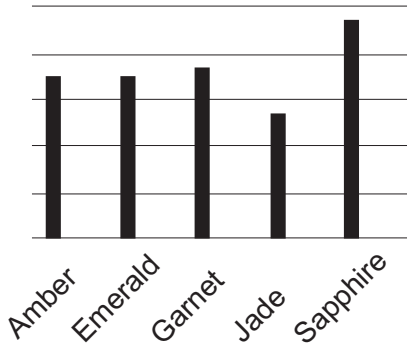
**A**



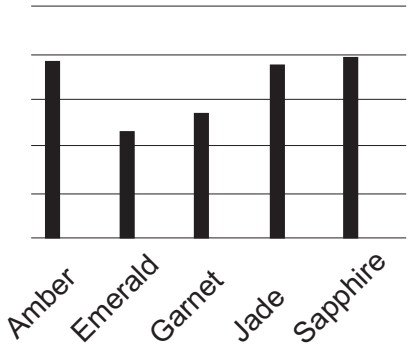

**B**



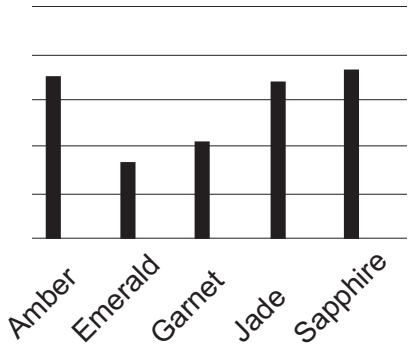

**C**




**D**

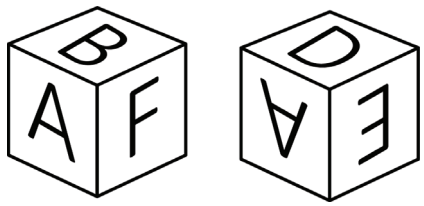



**E**



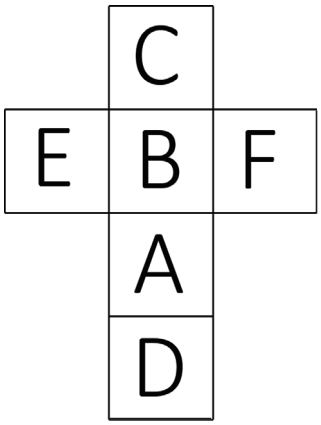

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**6** (2016 Q8) A cube is labelled with six letters and, when viewed in two positions, looks like this:

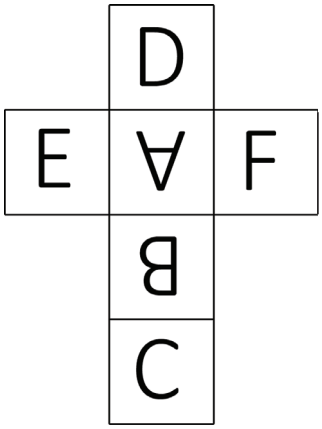



Which one of the following nets could be folded to make the above cube?

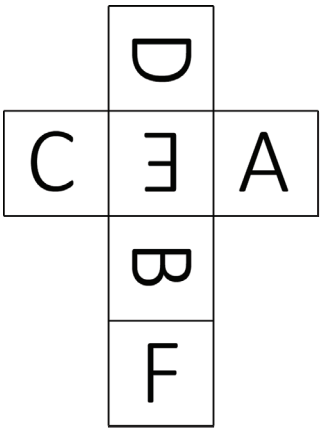
**A**



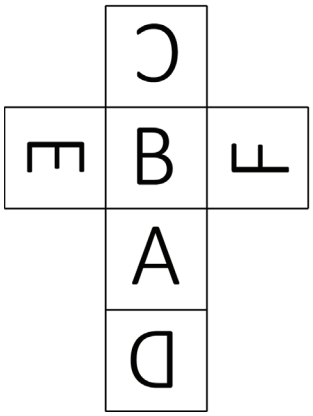

**B**



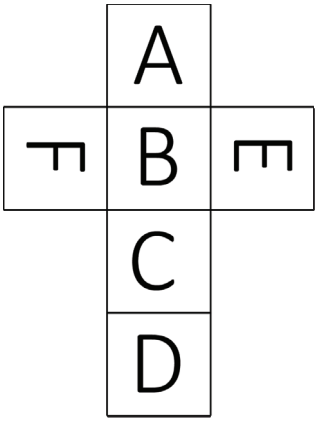

**C**




**D**

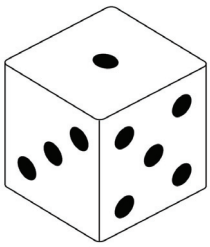



**E**



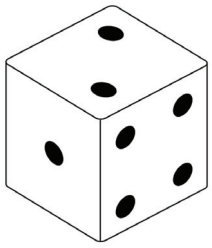

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**7** (2015 Q44) The die shown below is a conventional die, i.e. the total number of spots on any two opposite faces is always seven.

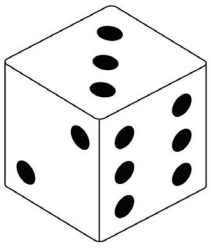


Which one of the following could **not** be the appearance of the die shown above after a subsequent throw?

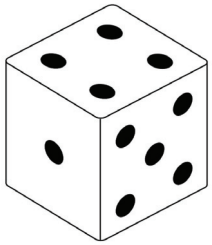
**A**



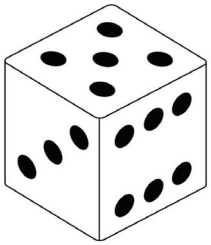
**B**



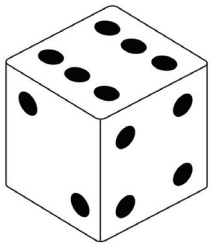
**C**



**D**

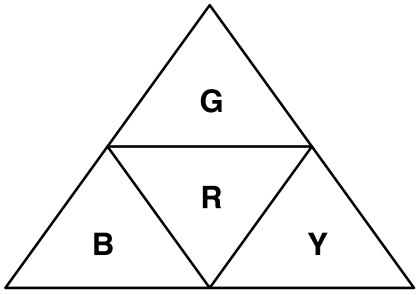


**E**



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**8** (2009 Q50) Four triangles of card coloured on one side are joined as shown.

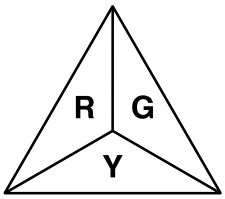



**G** = Green
**R** = Red
**B** = Blue
**Y** = Yellow

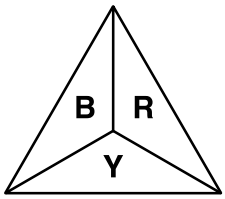
The joined triangles are then folded and joined to make a pyramid.

Which of the following could **not** be a view from above?

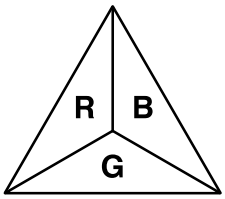
**A**



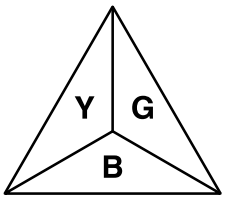

**B**



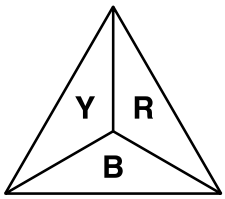

**C**




**D**

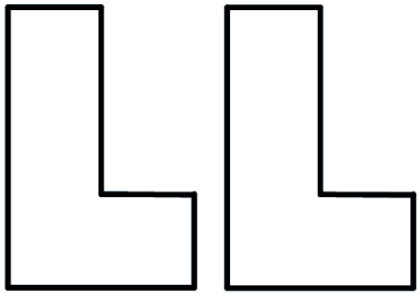



**E**




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**9** (2017 Q50) Marina purchased a pair of L-shaped tables thinking that they would add an unusual touch to her conference room.

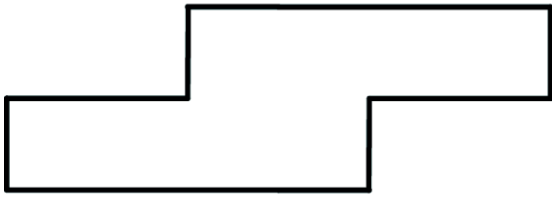


Which one of the following arrangements could **not** be made using the two L-shaped tables?

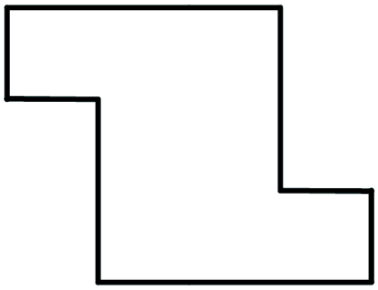
**A**



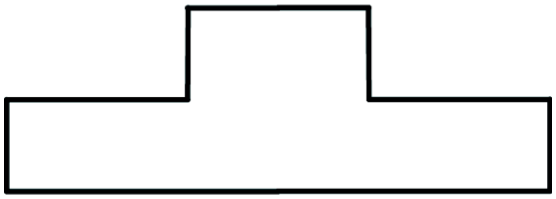
**B**



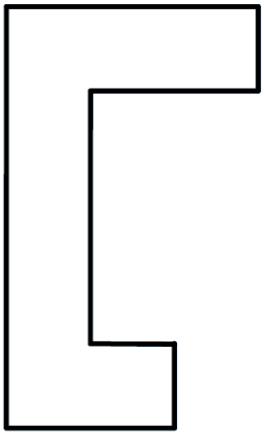
**C**



**D**

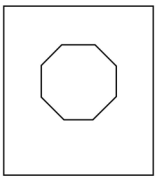


**E**



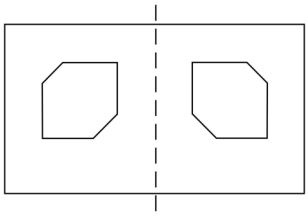
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**10** (2012 Q26) A sheet of paper has two holes cut from it. When it is folded in half the hole shown in the diagram below is seen.

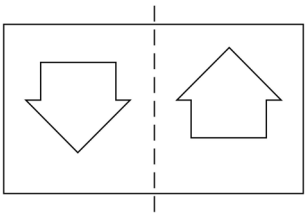


Which one of the following could **not** be the unfolded sheet of paper? (The dotted line indicates where it was folded.)

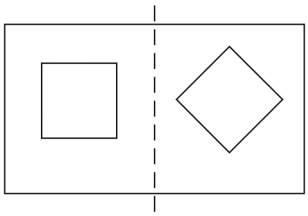
**A**



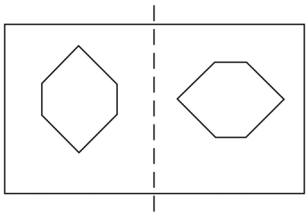
**B**



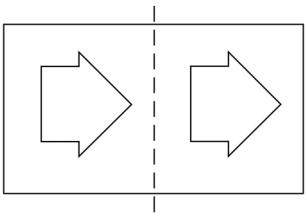
**C**



**D**

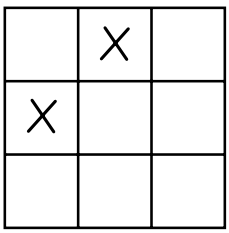


**E**



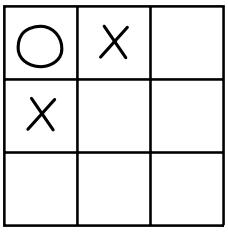
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**11** (2011 Q31) In a modified form of a popular game, each of two players at any turn may put either 'O' or 'X' as she chooses in any blank square of a 3 by 3 grid. She wins if she thereby completes a line of three identical symbols. A player is faced with this situation:

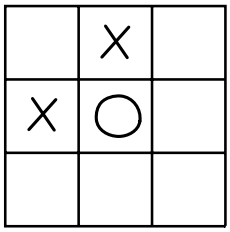


Which one of the following moves ensures that she can win at her next move?

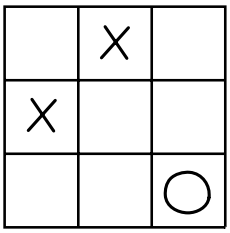
**A**



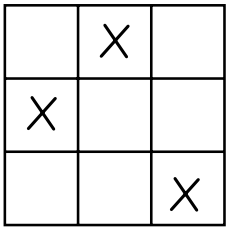
**B**



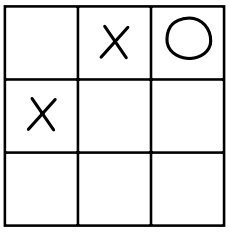
**C**



**D**



**E**



**12** (2009 Q48) Starting from its depot in town X, a delivery lorry is to make deliveries to three towns A, B and C, in any order which the driver chooses and finally to return to X.

Straight roads connect each town to every other town.

Depending on the layout of the towns, which of the following could **NOT** be the shortest journey for him to take?

**A** XAXBCBX

**B** XACBX

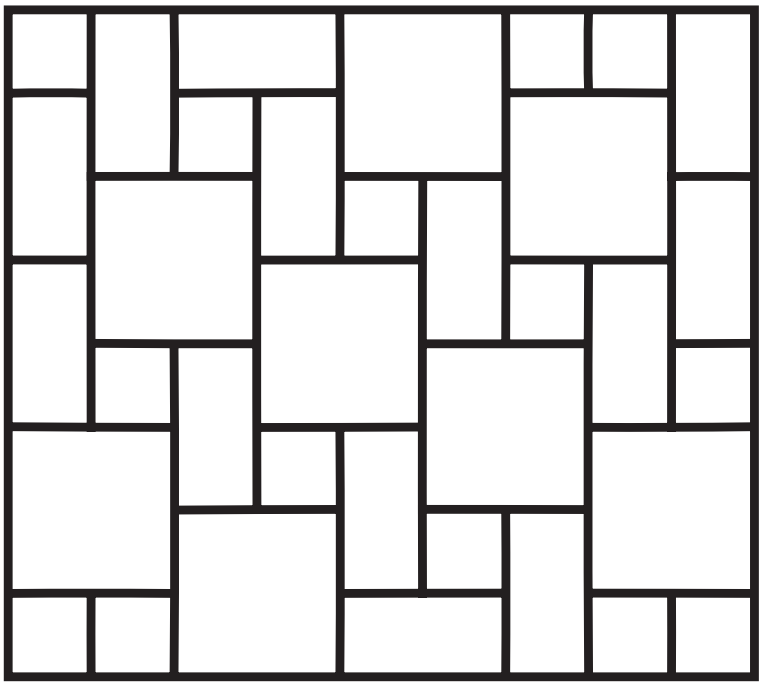
**C** XAXBCX

**D** XBABXCX

**E** XABCBAX

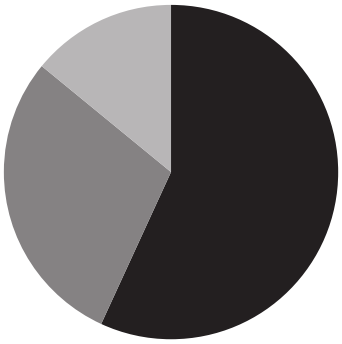
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**13** The block paved area shown below is laid using three different-sized blocks.
(2023 Q20)

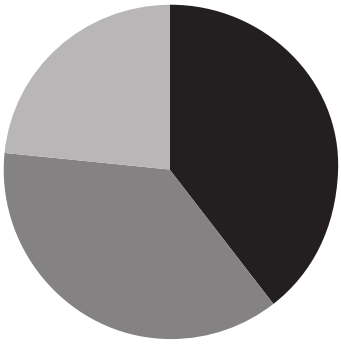


Which one of the following charts shows the relative comparison of the number of each size of block used in this paved area?

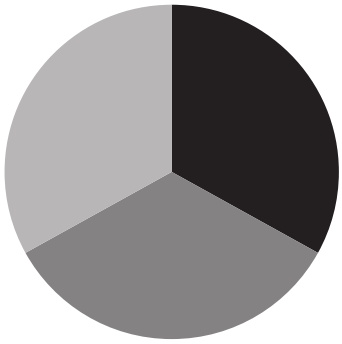
**A**



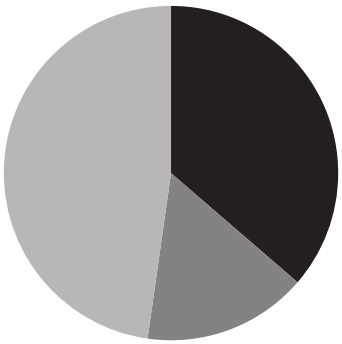
**B**



**C**



**D**



**E**



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**14** (2021 Q43) Jun lives in Applecross and Aki lives in Damsonville. Both live a 10-minute walk from their local train stations, which both lie on the Applecross–Figtown line.

Trains on this line run throughout the day to the following timetable:

| *Applecross–Figtown* | | | |
|---|---|---|---|
| *Applecross* | h:07 | h:27 | h:47 |
| *Bananagate* | h:17 | h:37 | h:57 |
| *Cherrytree* | h:25 | h:45 | (h+1):05 |
| *Damsonville* | h:34 | h:54 | (h+1):14 |
| *Elderhampton* | h:48 | (h+1):08 | (h+1):28 |
| *Figtown* | h:54 | (h+1):14 | (h+1):34 |

| *Figtown–Applecross* | | | |
|---|---|---|---|
| *Figtown* | h:07 | h:27 | h:47 |
| *Elderhampton* | h:13 | h:33 | h:53 |
| *Damsonville* | h:27 | h:47 | (h+1):07 |
| *Cherrytree* | h:36 | h:56 | (h+1):16 |
| *Bananagate* | h:44 | (h+1):04 | (h+1):24 |
| *Applecross* | h:54 | (h+1):14 | (h+1):34 |

"h" refers to all hours from 06:00 to 21:00. For instance, the train departing from Applecross at 08:27 arrives in Figtown at 09:14.

Jun leaves his house at 10:00 to go to visit Aki at her house. He must be back home by 16:00 and will spend at least 1 hour at Aki's house.

What is the difference between the maximum and minimum time that Jun can spend at Aki's house?

**A** 2 hours 40 minutes

**B** 2 hours 53 minutes

**C** 3 hours 26 minutes

**D** 3 hours 40 minutes

**E** 3 hours 53 minutes

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**15** (2021 Q31) Aliyah is planning a 2 week holiday in Spain with her friend Kyra and her husband. They have chosen their hotel and are calculating what the total cost will be for the 3 of them. They have decided on a departure date of May 21st and will be ready to make the booking in early January.

The following table shows dates and prices.

All prices shown are £s per person.

| *duration* | *Mar 29–Apr 22* | *Apr 23–29* | *Apr 30–May 6* | *May 7–13* | *May 14–June 10* | *June 11–17* | *June 18–24* |
|---|---|---|---|---|---|---|---|
| 7 nights | £580 | £600 | £650 | £670 | £700 | £700 | £700 |
| 14 nights | £900 | £900 | £970 | £1000 | £1050 | £1050 | – |
| 21 nights | £1200 | £1230 | £1300 | £1350 | £1400 | – | – |

Single room supplement: £11 on all dates.

Upgrade to a sea view room supplement: £3.

Supplement prices are per person, per night.

Early booking discount: if booked by 31st January is 5% up to a total of £2500, and 10% on any total above £2500.

Aliyah needs a single room. Kyra wants to upgrade the room for her and her husband so that it has a sea view.

What will be the total cost of the hotel booking for the 3 friends?

**A** £2181.20

**B** £2910.60

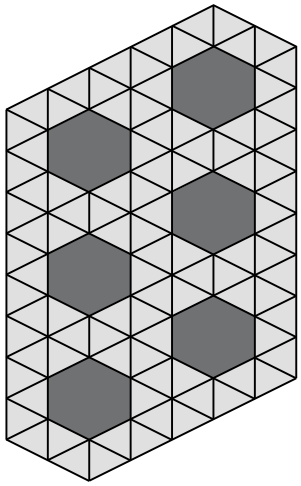
**C** £2973.60

**D** £3049.20

**E** £3218.60

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**16** (2009 Q44) The pattern below shows part of the tiling of a large floor in a community centre. It is made up of two types of tile: a hexagon and a triangle.



For a sufficiently large expanse of flooring, approximately what proportion of the two types of tile will be required?

**A** 1 hexagon to 6 triangles

**B** 1 hexagon to 9 triangles

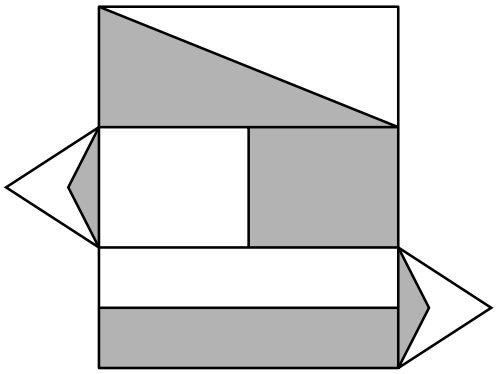
**C** 1 hexagon to 12 triangles

**D** 1 hexagon to 15 triangles

**E** 1 hexagon to 18 triangles

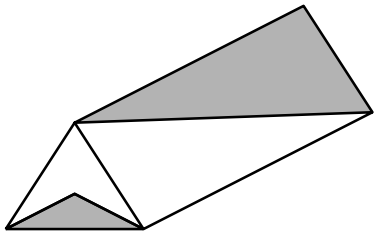
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**17** The piece of card shown below is folded to make a box with a triangular cross-section.
(2011 Q42)

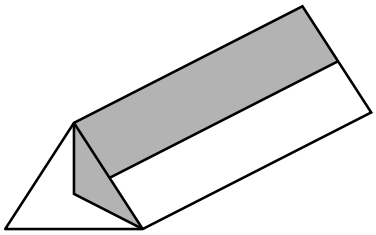


Which one of the views below can **not** be of this box?

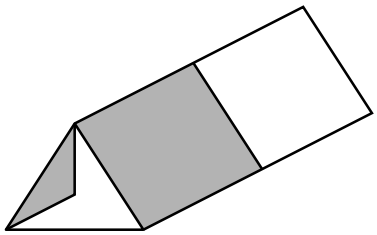
**A**



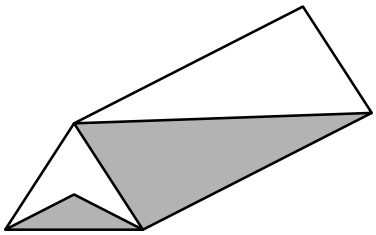
**B**



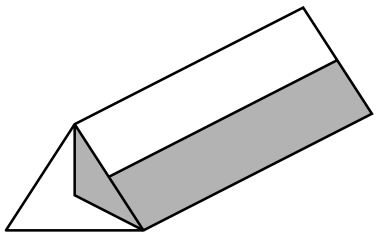
**C**



**D**

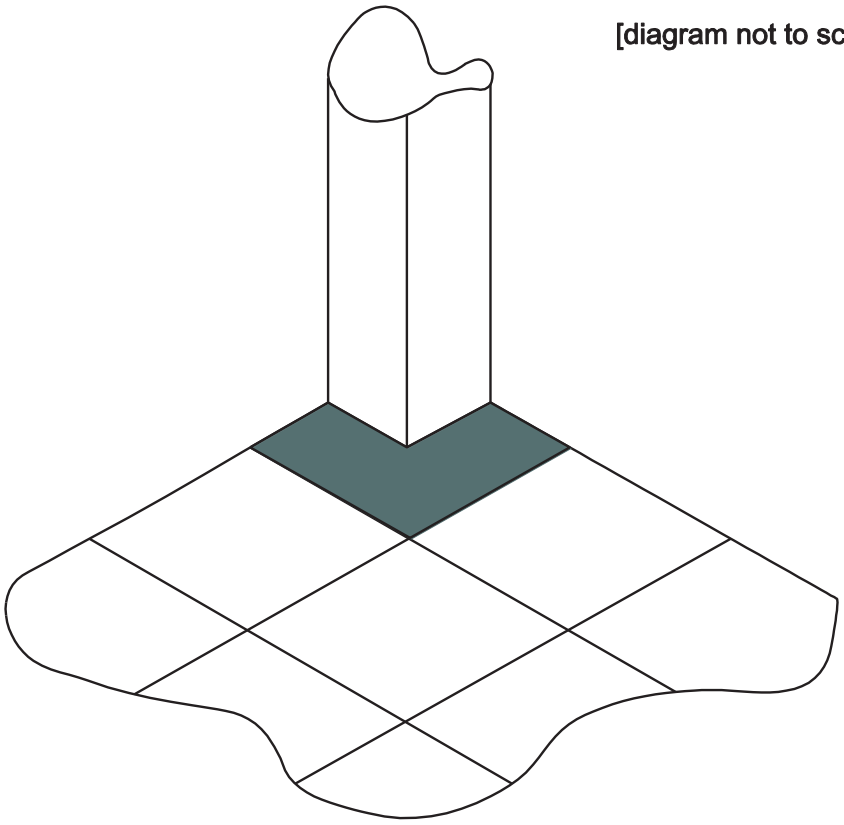


**E**

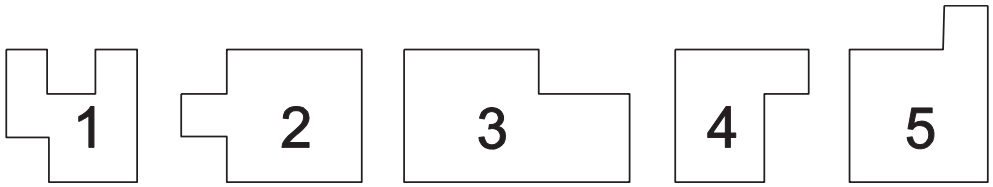


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**18** (2017 Q14) A tiler has encountered a problem when tiling a kitchen floor. He is one tile short of completing the job. This is indicated by the greyed-out area in the diagram below which shows part of the kitchen floor:




There are some odd pieces of tile left over that could be used to complete the job. Three of these, if combined, would fit the area exactly.




Which three pieces of tile could be pieced together to fit into the remaining corner area? (The tiles are shown face up and cannot be laid face down.)

**A** 1, 2 and 5

**B** 1, 3 and 5

**C** 1, 4 and 5

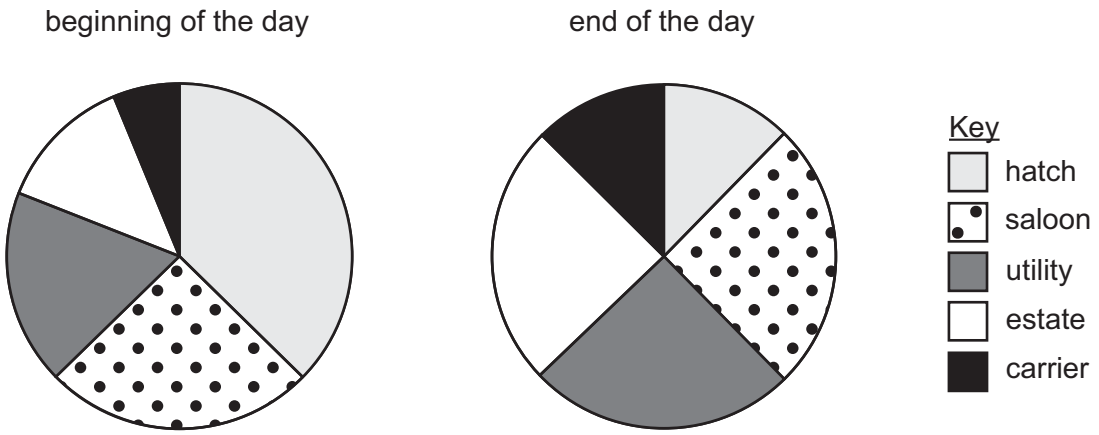
**D** 2, 3 and 4

**E** 2, 4 and 5

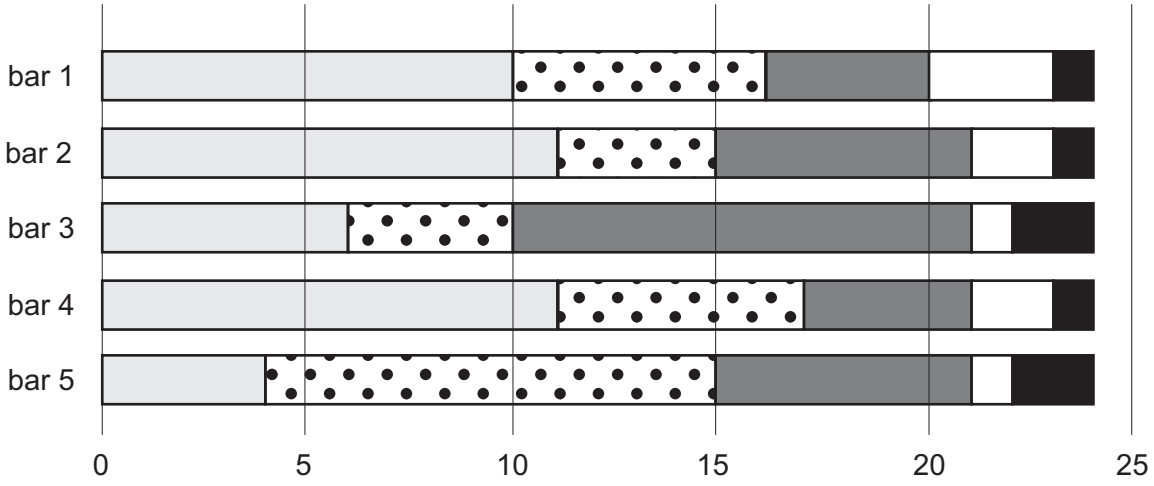
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**19** (2022 Q38) At the beginning of the day, a showroom had 32 vehicles for sale. The vehicles were advertised in five categories: hatchback, saloon, utility, estate car and people carrier. By the end of the day, 24 vehicles had been sold.

The first chart shows a comparison of the numbers of vehicles in each category in the showroom at the beginning of the day, and the second chart shows a comparison of the numbers in the same categories in the showroom at the end of the day.




Which one of the following shows a comparison of the numbers of cars in each category that were sold?




**A** bar 1

**B** bar 2

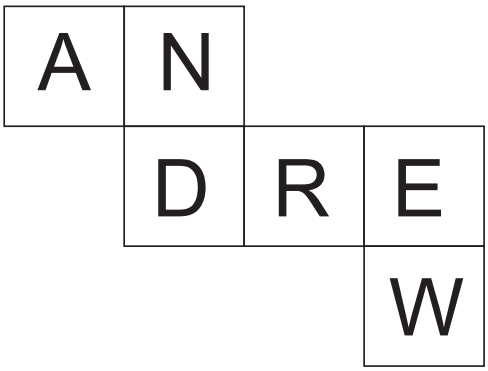
**C** bar 3

**D** bar 4

**E** bar 5

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**20** (2018 Q20) The diagram below shows the net of a cube.




The following diagrams show five suggested views of the cube once it has been assembled.




1 2 3 4 5

Which two of the above are possible views of the cube?

**A** 1 and 2

**B** 2 and 3

**C** 3 and 4

**D** 1 and 4

**E** 4 and 5

**21** (2014 Q36) Lucy allowed an hour to travel to a job interview, intending to arrive exactly on time. The train which she took arrived ten minutes late by her watch. The taxi which she then took got caught in a traffic jam for an unexpected 15 minutes and walking to the appropriate room took five minutes longer than anticipated. The clock in the interview room was running 30 minutes slower than Lucy's watch which was running ten minutes fast.

According to the clock in the interview room, when did Lucy arrive?

**A** She was ten minutes early.

**B** She was ten minutes late.

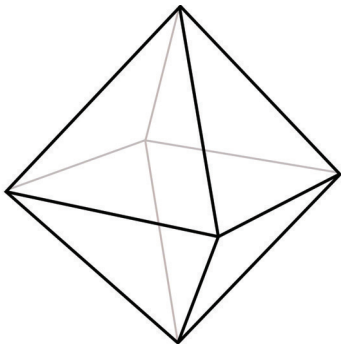
**C** She was thirty minutes late.

**D** She was sixty minutes late.

**E** She was on time.

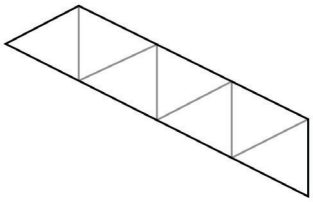
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**22** (2014 Q32) A regular octahedron is a solid composed of eight equilateral triangles, as shown in the image below:

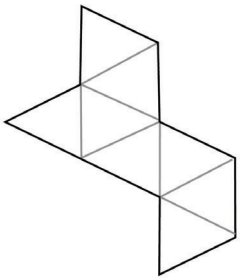


Which one of the following paper shapes will fold into an octahedron?

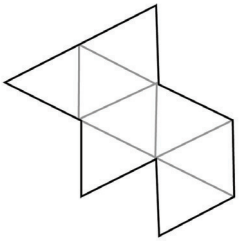
**A**



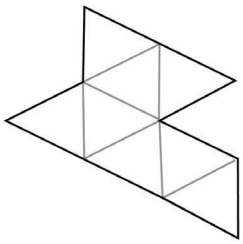
**B**



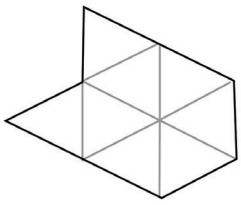
**C**



**D**



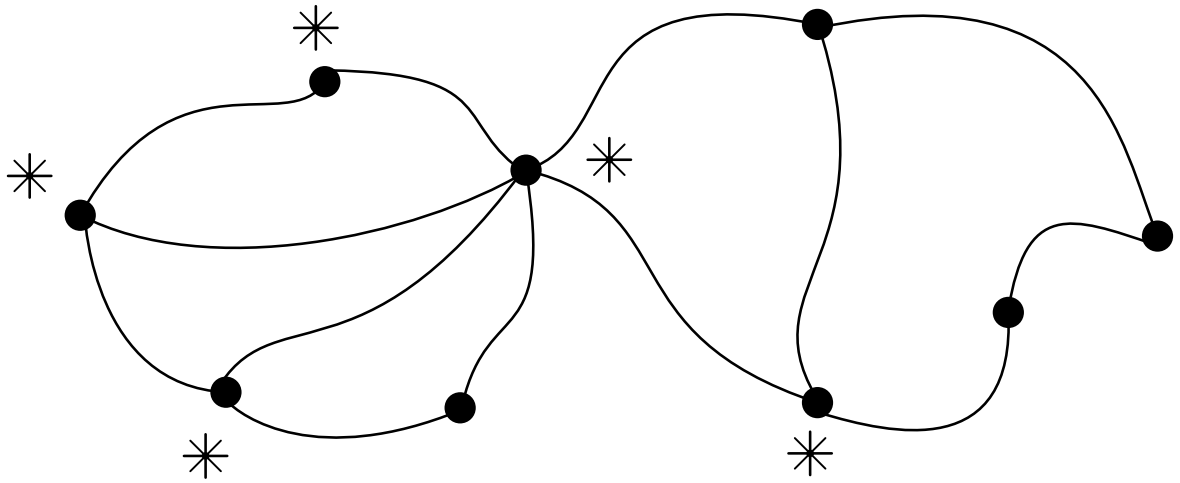
**E**



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**23** (2010 Q44)

My local park has a network of paths linking various locations, as shown below. The stars on the map represent five of the park's landmarks: the fountain, lake, bandstand, statue and bowling green.



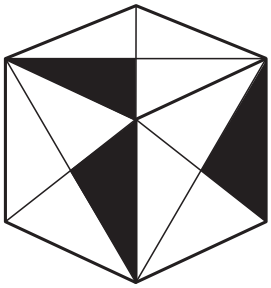
When I walk round the park, I often vary my routes and I do not always take the shortest route from one place to another. I have noticed that when I go from the statue to the fountain, I always pass the bandstand, but that going from the lake to the bowling green, I never pass the fountain unless I pass the bandstand twice.

How many of the stars on the diagram can be identified from the given information?

**A** 0
**B** 1
**C** 2
**D** 3
**E** 5

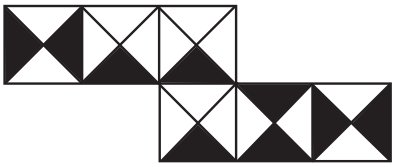
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**24** Here is a diagram of a cube.
(2018 Q44)

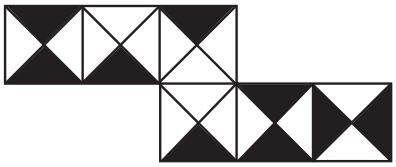


Which one of the following nets **could** be folded to make this cube?

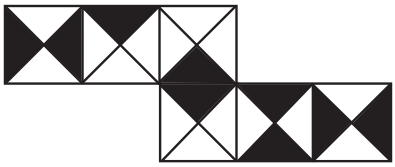
1



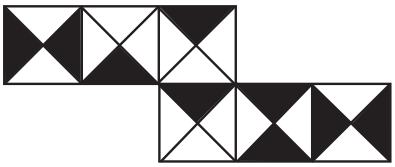
2



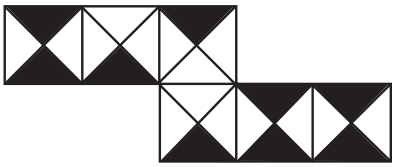
3



4



5



**A** 1

**B** 2

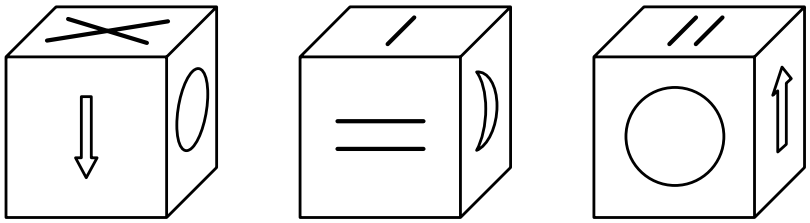
**C** 3

**D** 4

**E** 5

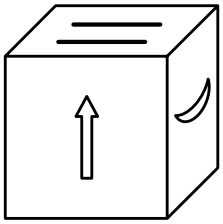
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**25** Three views of the same die with unusual symbols are shown below:
(2010 Q38)

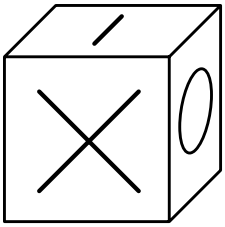


Which one of the following dice is **not** identical to the die above?

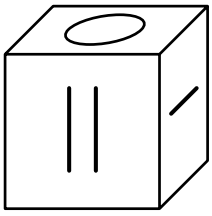
**A**



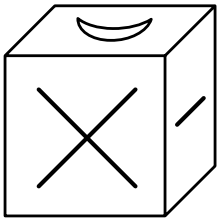
**B**



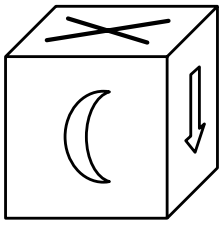
**C**



**D**



**E**



**END OF TEST**

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